

**Performance Work Statement
Dynamic Frequency Comb Applications Support
Indefinite Delivery, Indefinite Quantity (IDIQ)**

Requesting Laboratory / Division / Group:

Communications Technology Laboratory
Spectrum Research and Technology Division
Fiber Sources and Applications, (675.02)

I. BACKGROUND:

The Fiber Sources and Applications Group (675.02), at the National Institute of Standards and Technology (NIST) Boulder Colorado campus, develops novel applications of frequency combs for optical time transfer, spectroscopy and ranging. Of particular relevance here, this group has developed frequency comb-based optical two-way time-frequency transfer (O-TWTFT) that can allow frequency comparisons and time synchronization across a distributed network of clocks. They have also developed dual-comb spectroscopy for open-path atmospheric sensing. This work requires tight collaboration between laser science and digital control communities.

II. PURPOSE:

Frequency comb-based applications of time-transfer and dual-comb spectroscopy require advanced control algorithms and real-time processing. The objective of this Performance Work Statement (PWS) is to continue advancement of both systems by exploring the use of real-time phase/timing control of the frequency comb pulse train output. The work will develop the necessary digital-control of the phase/timing of the frequency comb output, contribute to the algorithm modifications and system modifications for both time-transfer and dual-comb spectroscopy to exploit this new capability, and then implement the new algorithms by modifying the existing NIST real-time digital signal processing systems accordingly.

This work will require the tight integration of FPGA based digital control systems with a number of laser frequency comb platforms. The goal is to build upon existing infrastructure in comb control including phase locking algorithms and FPGA platforms. Algorithm development by the contract will be performed with high level guidance only and the contractor will need to provide technical solutions to a range of scientific problems. Strong understanding of femtosecond laser operation, free-space optical time transfer and phase locking will be required.

The current state-of-the-art for this work can be found in the following publications
<https://www.nature.com/articles/s41586-022-05225-8> and
<https://www.nature.com/articles/s41586-023-06032-5>

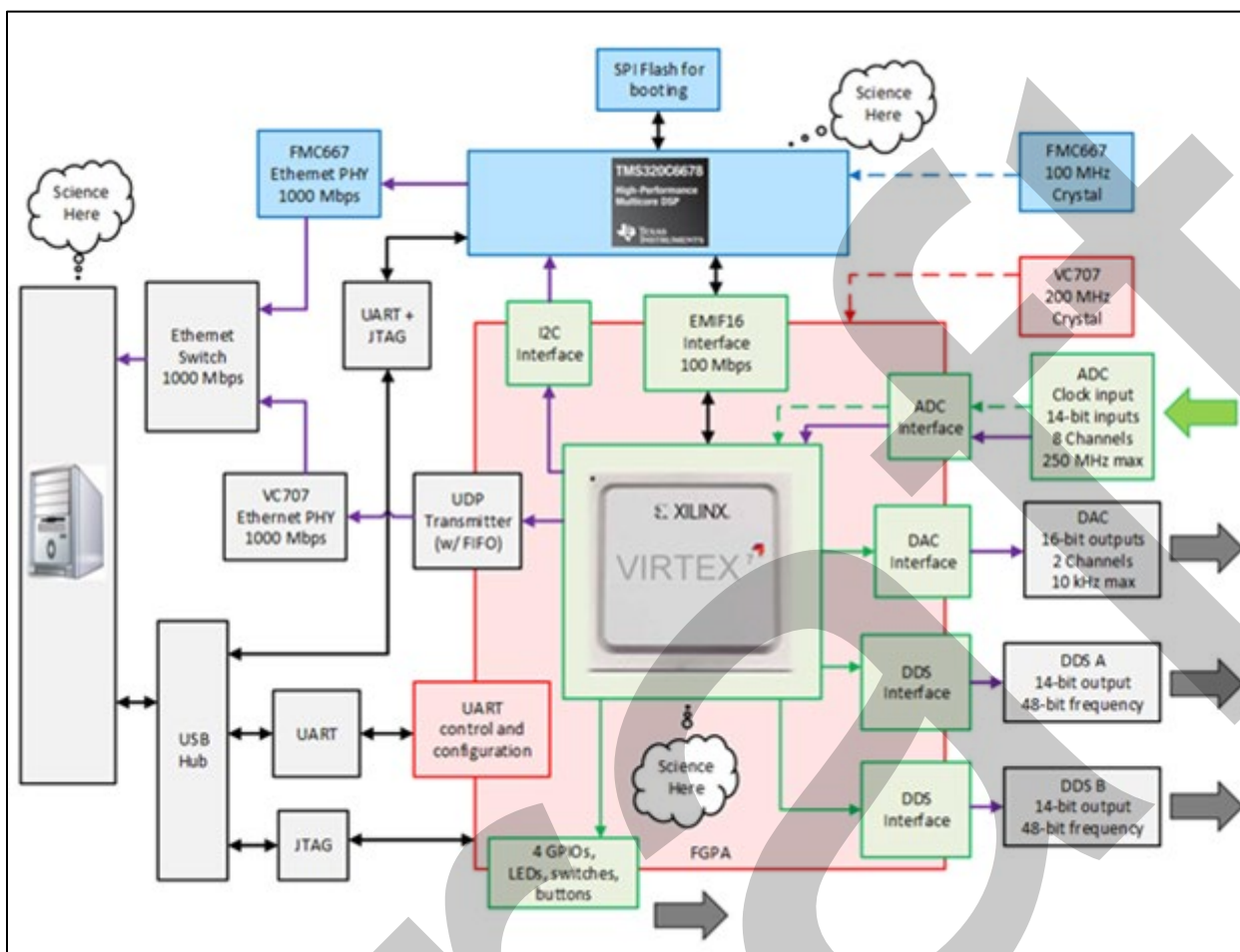


Figure 1: Existing real-time digital signal processor hardware for the optical time transfer. The digital signal processor for the dual-comb spectroscopy is similar to this design but uses a different FPGA. The digital control system for the frequency combs is located at <https://www.nist.gov/services-resources/software/fpga-based-digital-control-box-phase-stabilization-frequency-comb>.

Place of Performance:

The development work shall be completed at the Contractor's facility. Any required onsite testing and meetings will be conducted at the NIST, Boulder, CO campus, specifically Bldg. 1 (i.e., the Radio bldg.) and Bldg. 81 (i.e., the Precision Measurement's Lab). Specific travel requirements based on required tasks will be identified under the respective task order. The Contracting Officer's Representative (COR) and/or Subject Matter Expert (SME) may direct the flow of technical matters while on the Boulder campus.

Regular Business Hours:

Regular business hours are Monday through Friday, 8:00 am to 5:00 pm Local Time, excluding Federal holidays and NIST closures.

Period of Performance/Delivery Date:

Base Period	1 Year
Option Period 1	1 Year
Option Period 2	1 Year

Option Period 3
Option Period 4

1 Year
1 Year

III. SPECIFIC REQUIREMENTS:

1. The following are the types of services that may be procured via Task Orders:

- a. Task 1 - Algorithms and digital implementation for dynamic, arbitrary real-time control of the comb phase/time output

NIST frequency combs are controlled by a field-programmable gate array (FPGA) controller. Through real-time analysis and feedback to the comb, the controller is able to adjust the absolute timing of the comb pulses in an arbitrary fashion, limited only by the comb actuator. While the basic comb control has been shown, the Contractor shall build on existing technology to develop advanced algorithms to generate the maximum dynamic flexibility in the comb phase/timing control. The Contractor shall develop and implement algorithms necessary in VHDL/Verilog code for the FPGA. The Contractor shall modify this comb control to implement the applications.

Task 1.1 - Develop next generation firmware for maximum dynamic control of the frequency comb pulse train phase/timing to accommodate platform motion

The contractor will refine existing control firmware and algorithms to support new frequency comb platforms and actuator designs as well as scan patterns that will be required to accommodate platform motion and atmospheric induced timing noise in optical time transfer applications as well as novel scan patterns for spectroscopy. The firmware on the existing NIST FPGA comb controller shall be updated, tested and optimized for maximum dynamic control allowing the slewing of control signals to compensate doppler shifts.

Task 1.2 - Dynamic and coherent CW laser scanning

The contractor will propose and develop algorithms to maintain tight phase lock between a frequency comb and a CW laser while that laser is swept to match the expected Doppler shift profile associated with a reflection off of a LEO satellite. Sweep will match both the expected frequency shift range and slew rate while maintaining less than 0.1 rad of residual phase noise on the CW laser even in the presence of comb crossings.

Task 1.3 - Variable logging and system start up

Contractor will facilitate software and firmware development to ensure that critical measurement variables such as a repetition rate are implemented in the control software. Additionally, the contractor will contribute to the development of algorithms to simplify cold start operations. Contractor should propose a list of logged frequency comb parameters that will be necessary to recover all required time shifts. Logging and cold start algorithms will be updated with each iteration of the systems design on one year cadence.

- b. Task 2 - Long-range Optical Time Transfer through phase-controlled combs

Research has shown that Optical Time Transfer could operate at much lower received powers, and therefore longer range, if it were possible to dynamically control the phase of the frequency combs (see Task 1). The Contractor shall work with NIST to advance current algorithms to fully develop this

phase-control approach of the combs for a long-range mode of operation in the presence of platform motion. This task shall include contributions to overall system design, algorithm development, implementation in the existing NIST real-time digital signal processor through firmware modifications to both the FPGA and digital signal processor, field tests, and data analysis, as well as porting existing code bases to new FPGA platforms as old equipment becomes unavailable. This process shall involve the Contractor participating in at least two field tests of the system for a total duration of two months.

Task 2.1 - System design and algorithm development for platform motion

The Contractor shall develop tracking oscillator algorithms that can deal with the substantial platform motion expected when transferring femtosecond timing from ground to a LEO satellite. Work will mix algorithms for steering the frequency combs to ensure the pulses reflected off the satellite are always within 10 fs of the pulse center of the pulses from the local tracking oscillator comb. Additionally, the contractor should develop algorithms to ensure that the heterodyne beat between the Doppler-shifted return frequency comb and the tracking oscillator comb is well controlled enough to allow narrow filtering of return signals for improved signal in the presence of high link loss. Contractor should propose viable solutions for measuring and compensating for these Doppler shifts.

Task 2.2 - Implementation of Long-range Optical Time Transfer

The Contractor shall implement the algorithms by making modifications to the VHDL and C++ code in the existing real-time digital signal processor hardware for the optical time transfer. The Contractor shall also develop the Python code for the associated modifications to the GUI interfaces. The Contractor will also provide for the porting of existing capabilities to newer hardware platforms.

Task 2.3 - SLR station field tests and data analysis

The Contractor shall participate in three rounds of field tests of the optical time transfer system at an SLR station. Field tests are still being scheduled but will occur on a roughly 18-month cadence starting in September 2026. The Contractor shall perform data analysis and documentation of the results of the field tests and will work with the team to iterate on the control and Doppler correction algorithms to maintain femtosecond level time-transfer performance. The data analysis shall generate level 2 processed data directly from the raw data files saved by the real-time digital signal processor. The data analysis shall be completed in Matlab and the Matlab code library supplied to NIST with sufficient documentation to reproduce the analysis in another language such as Python.

Task 2.4 - SLR Optical Communication channel

The Contractor shall specify hardware, presumably an FPGA or RFSoc, and build a digital backbone for an optical communication channel on this platform. In the first iteration the digital backbone will generate a data stream at least 10 MB/s that will be written onto a cw laser with an EOM or used to directly amplitude modulate a diode laser. The second iteration will incorporate PRBS streams (or similar) that can be used for course ganging and timing measurements between nodes. A third iteration will push data rates up to 1 GB/s

Task 2.5 - Optical Terminal Tracking Control

The Contractor shall develop robust software solutions to interface with existing optical terminal hardware and will enable active tip tilt correction for the free space link at >100 Hz feedback bandwidths. Error readout in this is done with a SUI 320CSX camera and which will record the center of mass of the incoming beam and will feed a ± 2 V control signal for X and Y degrees of freedom to either a fast-steering mirror or a galvo mirror pair. Camera readout and center of mass processing as well as the generation of the control signal should be done on a microcontroller or the contractor's choosing.

c. Task 3 - Phase-stepped dual-comb

Dual-comb and related measurement techniques can operate with a much higher signal-to-noise ratio by dynamically controlling the phase of the frequency combs (see Task 1). The Contractor shall work with NIST to exploit this new capability by consulting on advanced algorithms and system design. The algorithms shall be capable of real-time operation on existing NIST digital processors. The system design shall be documented to the component level with a preference for standard components already in use by NIST. The Contractor shall modify the NIST FPGA comb controller system to interface with the dual-comb system both for control and data acquisition. The Contractor shall facilitate the testing of the system and testing with a focal plane array through algorithm development and firmware/software modifications. This process shall involve the Contractor participating in at least one lab experiment.

Task 3.1 Algorithms and system design for phase-stepped dual-comb spectroscopy

The Contractor shall develop the system design and algorithms for phase-stepped dual-comb spectroscopy based on the results of Task 1.

Task 3.2 Implementation and testing

The Contractor shall assist with the implementation of the necessary software/firmware changes in the existing NIST FPGA comb controller and dual-comb acquisition system for phase-stepped dual-comb spectroscopy.

For code developed under Tasks 1 through 3, the following specifications shall be met:

1. All code shall be covered with unit tests to at least 95% to verify functionality and to guard against regressions from future refactoring, extension, and/or bug fixes.
2. All code shall be supplemented by software test benches that enable simulation and unit testing without the need for hardware. Testing code may be written in Matlab or Python.
3. All code shall be documented by appropriate comments and naming conventions (in addition to Task 2 above.)
4. Proprietary or closed-source software, including FPGA IP cores, shall not be used unless the Contractor has previously verified NIST already owns such IP cores.
5. Code shall be organized to be reviewed by the Government on a regular basis, as specified in the deliverable table (Section IV). The results of the reviews are to be integrated into successive development steps.

6. This code shall be compatible with the existing firmware and custom NIST real-time processor for optical two-way time transfer.
7. All project software being developed shall be maintained in a suitable repository (e.g., GitHub) to allow Government to monitor development progress and provide rapid feedback when changes are needed.
8. The documentation shall include a description of the basic algorithms and their implementation in the DSP or FPGA. The documentation shall also include a basic mathematical description of the algorithm appropriate for publication in a scientific journal.

The code shall be implemented on the custom NIST real-time processor shown in Figure 1.

III. GENERAL REQUIREMENTS

1. Government Responsibilities:

Government will provide access to the time transfer systems during development cycles. Because of the cost of this access, access time will be limited to 2 weeks per year. Early-stage development should be done on a vendor owned test system and implemented on the government system only after significant testing has been done.

2. Government Furnished Property

All property, data and information provided by the Government in the performance of this task remains the property of the Government and shall be surrendered to the government upon completion or termination of this requirement. Likewise, all deliverables generated under this requirement remain the property of the Government.

IV. REQUIREMENTS FOR CONTRACTOR PERSONNEL

1. Homeland Security Presidential Directive-12 (HSPD-12)

The Contractor (and/or any subcontractor) and its employees who require physical access to NIST facilities or logical access to NIST IT networks or systems must comply with Homeland Security Presidential Directive (HSPD)-12, Policy for a Common Identification Standard for Federal Employees and Contractors; OMB M-05-24; OMB M-19-17; FIPS 201, Personal Identity Verification (PIV) of Federal Employees and Contractors; NIST HSPD-12 policy; and Executive Order 13467, Part 1 §1.2.

The Contractor (and/or any subcontractor) must submit a roster by name, position, e-mail address, phone number and responsibility, of all staff working under this acquisition where the Contractor will develop, have the ability to access, or host and/or maintain a government information system(s).

The roster must be submitted to the COR and/or CO within 14 days of the effective date of this contract. Any revisions to the roster as a result of staffing changes must be submitted within 14 days of the change. The COR will notify the Contractor of the appropriate level of investigation required for each staff member. If the employee is filling a new position, the Contractor must provide a position description, and the Government will determine the appropriate suitability level.

Position Sensitivity Designations Requirement

All Contractor (and/or any subcontractor) employees must obtain a background investigation commensurate with their position sensitivity designation that complies with Parts 1400 and 731 of Title 5, Code of Federal Regulations (CFR).

The following position sensitivity designation levels apply to this solicitation/contact:

Tier I

2. Safety: The Contractor employee must be responsible for knowing and complying with all installation safety prevention regulations. Such regulations include, but are not limited to, general safety, fire prevention, and waste disposal.
3. Security: NIST is a restricted campus. An identification badge is required for access into buildings and is shown to the armed Security Police when entering the campus.
4. Identification Badges: Contractor employees must comply with NIST identification and access requirements. Each Contractor employee must wear a visible identification badge provided by the NIST Security Office.
5. Vehicle Registration: All Contractor employees must register their vehicles with the NIST Security Office to gain access to the campus. A valid driver's license, Government-furnished civilian ID, proof of insurance and current registration must be presented to the NIST Security Office, at which time a NIST vehicle pass will be issued. The pass shall be displayed on the vehicle in accordance with NIST Security Office instructions.

V. SCHEDULE OF DELIVERABLES:

<i>Task Number/Reference</i>	<i>Description (Specific applicable tasks will be identified per task order)</i>	<i>Format</i>	<i>Quantity</i>	<i>Due Date (after start of Task Order)</i>
1.1	Develop next generation firmware for maximum dynamic control of the frequency comb pulse train phase/timing to accommodate platform motion	Code deposited to GitHub repository	1	12 months
1.2	Dynamic and coherent CW laser scanning	Code deposited to GitHub repository	2	9 months
1.3	Variable logging and system start up	Code deposited to GitHub repository	2	16 month
2.1	System design and algorithm development for platform motion	MS Word or PowerPoint	1	6 months
2.2	Implementation of Long-range Optical Time Transfer	Code deposited to GitHub repository	1	36 months
2.3	SLR station field tests and data analysis	Site visit for field tests	3	12 months

<i>Task Number/Reference</i>	<i>Description (Specific applicable tasks will be identified per task order)</i>	<i>Format</i>	<i>Quantity</i>	<i>Due Date (after start of Task Order)</i>
		Matlab code deposited to GitHub repository for analysis		
2.4	Optical Communication Channel	Code deposited to GitHub repository	2	12 months
2.5	Optical Terminal Tracking Control	Code deposited to GitHub repository	2	9 months
3.1	Phase-stepped dual-comb	Code deposited to GitHub repository	1	9 months
3.2	Implementation and testing	Code deposited to GitHub repository	1	6 months
1 to 2	Participate in weekly progress meeting with NIST staff either on-site or via NIST approved teleconference (e.g., Teams).	Hourly progress meeting	52	Weekly

VI. PERFORMANCE REQUIREMENT SUMMARY

<i>Desired Output</i>	<i>Required Service</i>	<i>Quality Standard</i>	<i>Monitoring Method</i>
Consulting Services in laboratory tasks and scientific research	Tasks 1 - 3	Research is of high scientific quality and conducted consistent with NIST, government and industry protocols, standards, and policies. Professionalism, quality, timeliness, and customer service are consistent with industry standards for the services provided. Services are timely, appropriate and responsive to Task Order requirements. Less than two deficiencies per period are observed.	Periodic inspection via TPOC or COR monitoring or customer feedback
Scientific data entry, log entries, and other technical written work	Tasks 1 - 3	Documents require less than one major revision. Government requests for clarification or correction are responded to within five business days. No incidences where responses are outside of 5 business days.	Documentation shall be reviewed monthly by the TPOC or COR.
Safety	Tasks 1 - 3	Requires 100% compliance with all safety requirements governed by Federal/State/Local law & regulation, NIST/unit policies, and lab procedures. At the start of performance of the first task order, contractor personnel will be	Periodic inspection by TPOC or COR

<i>Desired Output</i>	<i>Required Service</i>	<i>Quality Standard</i>	<i>Monitoring Method</i>
		provided NIST specific safety training totaling approximately 10 hours.	
Reports and presentations	Tasks 1 - 3	Clear and accurate and made in a professional manner. Draft presentations and reports submitted for approval in a timely manner 100% of the time. Less than two major revisions are necessary per contract period.	Documentation shall be reviewed monthly by the TPOC or COR
Software	Tasks 1-3	Performance requirement described in each task. Software must operate on intended platforms and in the intended environment without crashing more than once in an 80-hour period.	Software will be inspected within two weeks of delivery and will be expected to operate in a stable manner on the existing hardware and meet the program performance requirements.

VI. STANDARDS OF ACCEPTANCE:

The NIST Point of Contact (Technical Point of Contact or Contracting Officer's Representative) will provide comments on each deliverable within 3 weeks from receipt of a given deliverable. The Contractor shall make requested changes to the reports within 3 weeks from receipt of electronic or written comments from the NIST Point of Contact. The NIST Point of Contact will provide written notification when a deliverable is accepted.

VII. TRAVEL

It is anticipated the Contractor will be required to make at least three visits of one week or less to meet with NIST personnel either in Boulder, CO or at another meeting location per year. Specific travel requirements will be identified per Task Order. Travel costs will adhere to federal travel regulations, <https://www.govinfo.gov/content/pkg/CFR-2021-title41-vol4/pdf/CFR-2021-title41-vol4.pdf>.

VIII. CONTRACTOR'S MINIMUM QUALIFICATIONS

1. Contractor shall serve as the Key Personnel position "Algorithm and Signal Processing Lead" on the project.
2. Shall have staff with a Ph.D. in physics or electrical engineering
3. Shall have over eight years' experience in engineering embedded systems hardware and firmware, digital signal processing.
4. Shall have expertise in multiple areas of signal processing including communications, frequency/time generation, and transfer, analog and digital electronic design, as demonstrated by past work experience.

5. Shall have expertise in VHDL, Verilog, C, C++, Python, Matlab
6. Shall have demonstrated substantial experience with and successful prior independent implementation of:
 - a. Electro-optic systems with femtosecond timing jitter, such as fiber laser frequency combs or other related technologies.
 - b. Complete embedded system development including development of system requirements, hardware design, build, testing, and code development. Systems should include use of FPGA, DSP, Ethernet, serial peripherals,
 - c. Construction of electro optic systems and firmware for the implementation of optical communications over free space.
 - d. Development of embedded digital electronics products for basic science research
7. Shall have specific expertise in building digital phase lock systems for laser phase locking. The contractor should have experience implementing multiple simultaneous phase locks on a single FPGA with lock bandwidths greater than 200 kHz. Contractor should have experience with both frequency-to-cw laser locks and frequency comb carrier-envelope-offset frequency locks.
8. Shall have experience working closely with physical scientists as end customers. References shall be provided from previous basic science collaboration work, attesting to the qualification of the Contractor for a project of this nature and the ability to communicate with physical scientists.

Or, a demonstrated equivalent combination of education, training, and experience can be used to meet the minimum qualifications. This person shall be able to perform, document, and improve difficult science and engineering tasks. This person typically has recognized competence through a track record of technical contribution; experience with solving complex technical problems; and effective written and oral English communications for the specific task.

IX. CONFIDENTIALITY

The information developed under this contract is the property of the US Government and shall be kept in strict confidence.